



Common hazards

Some of the hazards of the jet age are as old as flying itself. In October 1960, for example, a Lockheed Electra turbojet taking off from Boston flew into a dense swarm of starlings, which brought it down, killing 62 people. Slush on runways has been the cause of a number of notorious crashes, including the Munich air disaster that killed so many of the young Manchester United football team in 1958. Flying by instruments has taken much of the sting out of low cloud and fog, but it is still safer to fly when you can see where you are going. This is especially true around the more hazardous airports – pilots do not look forward to instrument landing where air-traffic control is not of the highest standard and flight paths bring the aircraft in low over or between mountains or hills.

Aircraft still need to avoid thunderstorms or typhoons, although improved weather forecasts and onboard weather radar have made this far

easier to achieve. Violent winds, notably the downward waves of air sometimes encountered in the lee of mountains, can still be treacherous for modern planes – one such caused a Uruguayan aircraft to come down in the Andes in October 1972, a crash that became notorious when it was later revealed that the survivors had resorted to every means possible to stay alive. Wind shear, a sudden change in wind direction or velocity that occurs where different bands of air meet in the jet stream, can cause airliners to drop vertically for hundreds of feet, a terrifying and potentially fatal experience for passengers.

Midair collisions

As has been mentioned, the first midair collision between airliners happened in 1922. Fears about such events are stoked by scary statistics about the number of reported close encounters between aircraft around busy major airports. The world's worst ever midair collision, between a Saudi Boeing 747 and a Kazakh Airways Ilyushin Il-76, outside Delhi, India, in 1996, killed 350 people. But even if something has gone wrong, it requires a lot of bad luck to actually hit another moving object in what is actually quite a large three-dimensional space. Tragic incidents from the propliner era, such as the collision of a United Airlines DC-7 and a TWA Constellation over the Grand Canyon in 1956, and of a TWA Super Constellation and a United DC-8 over Staten Island in 1960, were examples both of flaws in air-traffic control and of extreme ill fortune. The presence of private aircraft in the same airspace as

TRAGIC COLLISION

A Boeing 727 with 135 people on board plunges in flames after colliding with a Cessna light aircraft over San Diego, California, in September 1978. It was a clear day and the Cessna pilot was operating on Visual Flight Rules. All on board the 727 were killed, plus two people in the Cessna and two on the ground.

commercial jets creates the greatest potential for collisions – as was shown in September 1978, when a Pacific Southwest Airlines 727 collided with a much slower light aircraft over San Diego, an accident that cost more than 150 lives.

Mechanical faults

Serious flaws in aircraft can elude the usually rigorous testing procedures of manufacturers and regulatory authorities. In 1959–60, for example, two of the newly introduced Lockheed Model 188 Electra turbojets crashed through a wing falling off as a result of a weakness in the engine nacelles. America's biggest single air accident, a DC-10 crash at Chicago in May 1979, which killed all 271 people on board, was caused by the failure of a bolt, leading to the separation of an engine from the pylon assembly – all



RIVER CRASH-LANDING

Air-accident investigators often have to work under difficult conditions. This aircraft operated by Indonesia's Garuda airline crash-landed in the Bengawan Solo river in Klaten, Central Java, in January 2002. Indonesian investigators can be seen inspecting the wreckage and searching for debris in the river.

